

Redrob Candidate Ranker

India Runs: Data & AI Challenge

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Problem Statement: Legacy applicant tracking systems rely on rigid keyword matching, missing perfect candidates who use different terminology. Our solution aims to solve this by building a highly intelligent, 5-stage semantic AI.

Solution Overview

What is your proposed solution?

An elite, "God-Tier" Omni-Architecture utilizing a Two-Stage NLP pipeline (Bi-Encoder + Cross-Encoder) combined with psychological profiling and heuristic behavioral extraction.

What differentiates your approach?

1. **Psychological Synthesis:** We automatically synthesize a proxy for the candidate's "Big Five" (OCEAN) personality traits using open-source heuristic formulas.
2. **Two-Stage NLP:** A lightning-fast HuggingFace Bi-Encoder finds the top 1,000, then a powerful Microsoft MARCO Cross-Encoder perfectly re-ranks the top 200.
3. **Action-Oriented Leadership Detection:** Our linguistic engine scans resumes for active leadership verbs to automatically detect "Doers" vs "Participants".

JD Understanding & Candidate Evaluation

What are the key requirements extracted?

Core AI competencies (embeddings, retrieval, python), product-company experience (vs pure service), and minimum 3 years of experience.

Which candidate signals are most important?

We evaluate candidates on 5 unique dimensions:

1. **Semantic Match:** Deep contextual similarity to the JD.
2. **Career Stability:** Mathematically penalizing job hopppers.
3. **Leadership Bias:** Tracking their use of action verbs.
4. **Modesty (Dunning-Kruger):** Penalizing juniors who claim 20+ "Expert" skills.
5. **Promotion Velocity:** Rewarding those fast-tracked to Senior roles internally.

Ranking Methodology

How does your system retrieve, score, and rank?

1. **Hard Filter:** Drops candidates with `<3` YoE or honeypot traps.
2. **Scoring:** BM25 mixed with Behavioral, Geo-Viability, and Learner Multipliers.
3. **Bi-Encoder Re-Ranking:** Top 1,000 scored via Cosine Similarity (`all-MiniLM-L6-v2`).
4. **Cross-Encoder Re-Ranking:** Top 200 run through a deep-interaction Cross-Encoder.
5. **Reciprocal Rank Fusion (RRF):** Perfect mathematical aggregation of discrete ranks.

Explainability & Data Validation

How are ranking decisions explained?

Our open-source system features a fast, rule-based NLP engine that instantly generates a 3-paragraph **Executive Summary** mimicking a Generative LLM, explaining why the candidate is the perfect hire.

How do you prevent hallucinations?

No Generative AI is used at runtime. All reasoning, skill consistency checks, and personality scores are extracted purely deterministically from the candidate's verified metadata.

End-to-End Workflow

1. **Ingest:** Stream massive `candidates.jsonl` files iteratively.
2. **Filter & Infer:** Discard honeypots and infer missing latent skills dynamically.
3. **Expand & Retrieve:** Run BM25 with AI Knowledge Graph Query Expansion.
4. **Two-Stage NLP:** Run the Bi-Encoder -> Cross-Encoder dual architecture.
5. **Psychological Profiling:** Synthesize the Big Five personality traits and Leadership score.
6. **Output:** Generate `team_PJ2001_IND.csv` with injected X-Ray JSON metadata.
7. **Visualization:** Our custom Hugging Face UI instantly renders Interactive 3D Radar Charts and Knowledge Graphs from the CSV.

System Architecture

- **Data Layer:** Streaming JSONL parser.
- **Retrieval Engine:** Custom BM25 implementation with Latent Skill Inference.
- **NLP Engine:** HuggingFace `sentence-transformers` (`all-MiniLM-L6-v2` + `ms-marco`).
- **Aggregation Layer:** Reciprocal Rank Fusion (RRF) algorithm.
- **Presentation Layer:** A stunning Streamlit UI built with `plotly` (Radar Charts) and `graphviz` (Skill Knowledge Networks).

(Everything runs 100% locally. No closed-source external APIs are used.)

Results & Performance

What results demonstrate ranking quality?

The system successfully identified highly-qualified Senior Engineers who possessed the required skills but suffered from "resume modesty" by automatically inferring their latent skills and rewarding their high career velocity.

How does it meet compute constraints?

- **Scale:** Successfully evaluated the entire 100,000 candidate dataset.
- **Speed:** Even while computing Cross-Encoders, Temporal Consistency, Personality traits, and NLP summaries, the total end-to-end runtime is **~23 seconds** on standard CPU.
- **Memory:** Stays well below 16GB due to efficient streaming pipelines.

Technologies Used

- **Python (3.11+)**: Core pipeline architecture.
- **Sentence-Transformers**: For open-source, CPU-optimized NLP models.
- **Plotly & Graphviz**: For generating 3D interactive Radar Charts and spider-web Knowledge Graphs natively in the UI.
- **Scikit-Learn**: For rapid matrix calculations.
- **Streamlit**: For hosting the interactive "X-Ray Vision" God-Tier dashboard.
- **Hugging Face Spaces**: Used to host the live Sandbox environment.

Submission Assets

- **Live Sandbox Demo:** <https://huggingface.co/spaces/praasukjain2001/redrob-ranker>
- **GitHub Repository:** <https://github.com/PJ2001-IND/redrob-ranker>
- **Output File:** `team_PJ2001_IND.csv` (and `.xlsx`)
- **Metadata:** `submission_metadata.yaml`



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